

Lecture 5

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Programming Languages (PL)



Language

- ◆ Syntax
- ◆ Semantics

Syntax

- ◆ Written aspect of language
 - ◆ Rules
 - ◆ Convention

Semantics

- ◆ The meaning of language

“The Meaning of Language”

- Savage, Ebberts, Martin

“If people do not believe that mathematics is simple, it is only because they do not realize how complicated life is.”

—John von Neumann

Python

Python

- ◆ Syntax
 - ◆ White space based
 - ◆ Indentation

Python Binding

name = value

Value is typed based on inference.

Python function

```
1 def function_name(parameters, x, y):  
2     # indent  
3     # logic here  
4     c = x + y  
5     return c
```

$f(x, y, \dots) < colon >$
 $< indent > logic$

Python Iteration

```
1 for i in S:  
2     print(i)
```

for < iterator > *in* < collection > < colon >

< logic >

=

$\forall$ < iterator > $\in$ < collection >

< logic >

“The more I think about language, the more it amazes me that people ever understand each other at all.”

-Kurt Gödel

Code Conventions

- ◆ A set of rules to constrain your programming
- ◆ Etiquette

Documentation

- ◆ Internal
- ◆ External
- ◆ Both

“There is a difference between a thing and
talking about a thing.”

-Kurt Gödel